

Comparative Accuracy of TCD, TTE, TEE, and Cardiac CT in Detecting Right-to-Left Shunt in Embolic Stroke of Undetermined Source

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Abstract

Background and Objectives

Identifying a right-to-left shunt (RLS), commonly due to a patent foramen ovale (PFO), is essential in evaluating embolic stroke of undetermined source (ESUS). While several imaging modalities are used for PFO detection, direct comparisons in a single cohort are limited. This study evaluates and compares the diagnostic performance of 4 major modalities: transcranial Doppler (TCD), transthoracic echocardiography (TTE), transesophageal echocardiography (TEE), and cardiac CT in patients with ESUS.

Methods

We retrospectively identified patients with ESUS referred to our neurosonology laboratory from July 2023 to May 2025, who underwent at least 3 of the 4 diagnostic modalities. A confirmed PFO was defined by concordant positive findings on at least 2 tests. Diagnostic performance metrics (sensitivity, specificity, positive predictive value, negative predictive value [NPV], accuracy, and area under the curve [AUC]) were calculated, with pairwise comparisons assessed using the DeLong test.

Results

Of 469 screened patients, 130 met inclusion criteria. TCD showed the highest sensitivity (100%), NPV (100%), and an accuracy of 87.6%, although specificity was moderate at 75.0%. After excluding patients with mild PFO or extracardiac shunts, specificity and accuracy improved to 81% and 90.8%, respectively. TTE achieved a sensitivity of 83.3% and specificity of 93.3%, with an overall accuracy of 88.1%. TEE had a sensitivity of 95.1%, specificity of 87.5%, and the highest overall accuracy at 91.3%. Cardiac CT had perfect specificity (100%) but the lowest sensitivity (45.8%), limiting its standalone diagnostic utility. AUC comparisons revealed superior diagnostic performance for TCD, TTE, and TEE over cardiac CT (p values <0.05), but no significant differences among TCD, TTE, and TEE.

Discussion

TCD is a highly sensitive, noninvasive first-line tool for detecting RLS in ESUS, particularly in acute or resource-limited settings. Confirmatory TTE, TEE, or CT can be tailored to context, with a tiered approach optimizing PFO detection and prevention strategies. Results should be interpreted cautiously, given the referral and selection bias. Larger prospective multicenter studies are needed to validate this approach and improve outcomes.

Classification of Evidence

This study provides Class IV evidence that (1) TCD and TEE were highly sensitive in detecting PFO; (2) TTE and cardiac CT were highly specific in detecting PFO; (3) TCD, TTE, and TEE had superior diagnostic performance over cardiac CT (as defined by AUC comparison analysis); and (4) there were no significant differences in overall diagnostic performance among TCD, TTE, and TEE.

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Supplementary Material

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Introduction

Embolic stroke of undetermined source (ESUS) accounts for 7–42% of all ischemic strokes, representing a significant diagnostic and therapeutic challenge in stroke medicine.^{1,2} ESUS is generally defined by the absence of a large artery atherosclerotic source, major cardioembolic source (e.g., atrial fibrillation [AFib]), or small vessel disease etiology.^{1,2} Despite rigorous diagnostic workup, including prolonged cardiac monitoring, neuroimaging, and vascular studies, a large proportion of ischemic strokes remain unexplained, underscoring the complexity and heterogeneity of ESUS.³

A key mechanism implicated in ESUS involves paradoxical embolism through a patent foramen ovale (PFO), a congenital, flap-like communication between the right and left atria that fails to close postnatally.⁴ Recent estimates suggest that PFO may be involved in approximately 40% of ESUS cases, making its detection a high clinical priority.^{2,5} Recognition of the potential link between PFO and cryptogenic stroke has motivated investigations into whether PFO closure can reduce recurrent stroke risk, with randomized trials and observational studies demonstrating potential benefits in carefully selected patients.^{4–7} However, the decision to proceed with invasive closure depends heavily on accurate PFO detection because an incorrect diagnosis can lead to unnecessary procedures or missed therapeutic opportunities.

Multiple diagnostic modalities are available for PFO detection, each with unique strengths and limitations. Transcranial Doppler (TCD) with bubble study detects microembolic signals in cerebral vessels after intravenous injection of agitated saline and is known for its high sensitivity but lower specificity and inability to visualize cardiac anatomy.⁸ Transthoracic echocardiography (TTE) with bubble study is broadly accessible but dependent on acoustic window quality and Valsalva maneuver effectiveness.⁹ Transesophageal echocardiography (TEE), although often considered the reference standard for anatomical detail, is invasive and subject to patient intolerance and operator variability.¹⁰ Cardiac CT offers high-resolution structural imaging but cannot assess dynamic shunting under physiologic conditions.¹¹

Recent studies highlight that no single imaging modality is sufficient for comprehensive detection of right-to-left shunt (RLS) or PFO. TCD offers the highest sensitivity for initial screening, up to 96%, outperforming both TTE and TEE, particularly when combined with a Valsalva maneuver.^{12,13} However, TCD may have a higher false-positive rate, as noted by published studies which found TTE to be more specific in their comparative analysis.¹⁴

While earlier studies have primarily focused on comparing combinations of TCD, TTE, and TEE—often omitting cardiac CT—our study takes a more comprehensive and clinically relevant approach by evaluating all 4 imaging modalities within the same cohort of patients with ESUS. This broader analysis is especially timely, as cardiac CT has seen increasing use in stroke

evaluations according to recent reports, whereas TCD remains relatively underused in many US centers, which is an observation based on our institutional and anecdotal experience, despite its well-documented diagnostic strengths.

The aim of this study was to directly compare the diagnostic performance of TCD, TTE, TEE, and cardiac CT for the detection of PFO in patients with ESUS. The study focuses on identifying the imaging modalities with the highest sensitivity and specificity for detecting PFO, comparing the overall diagnostic accuracies of TCD, TTE, TEE, and cardiac CT including assessment by area under the curve (AUC), and evaluating potential differences in performance among TCD, TTE, and TEE within the same patient cohort.

Methods

Study Design and Population

This study was conducted as a retrospective analysis at a single tertiary care center. The cohort included consecutive patients aged 18 years or older who were referred to our center for workup of ESUS between July 2023 and May 2025. ESUS was defined based on standard criteria, including an acute ischemic stroke with no evident large artery atherosclerosis, no high-risk cardioembolic source, and an absence of other specific etiologies, such as small vessel disease or hypercoagulable states.¹⁵ All patients had ischemic stroke confirmed by neuroimaging. In addition, their imaging showed no significant arterial blockages or identifiable sources of embolism such as dissection. Patients with a history of AFib were included if their current ischemic event was a recurrent stroke that occurred despite best medical therapy (including anticoagulation) for their AFib, suggesting that AFib was not the sole or definitive cause of the index stroke. These patients were specifically investigated for an alternative mechanism, such as RLS. We excluded patients with an active hypercoagulable state (e.g., symptomatic deep vein thrombosis/ pulmonary embolism or new genetic hypercoagulability requiring acute/recent management) at the time of the index stroke. Patients with a history of hypercoagulable state were included, as their index stroke was still considered undetermined based on the remaining workup.

The RLS was localized using agitated saline contrast (bubble study). Intracardiac shunt was defined by the direct visualization of bubbles crossing the interatrial septum into the left atrium within 1–3 cardiac cycles in TEE studies. Extracardiac shunts were defined by the delayed appearance of bubbles in the left atrium (≥ 4 to 6 cardiac cycles) or clear evidence of intrapulmonary shunting. In patients evaluated with TCD, a delayed microbubble signal in the middle cerebral artery (typically appearing ≥ 30 seconds after injection) was another supportive clue for an extracardiac source.

All patients deemed to have a high clinical suspicion for PFO underwent at least one of the 4 examinations; however, only

those who had undergone at least 3 of these diagnostic tests, TCD, TTE, TEE, or cardiac CT, were included in the final study analysis. Details for each diagnostic procedure can be found in eMethods.

Statistical Analysis

A true-positive PFO case was defined as being positive in at least 2 different imaging modalities. Patient demographics, comorbidities, and imaging findings were collected from electronic medical records. The sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and accuracy for each modality were calculated based on whether PFO was confirmed (true positive) or rejected (true negative) by at least 2 modalities. A PFO detected by 1 modality but not confirmed by a second test was considered a false positive. Conversely, if the PFO was positive in at least 2 modalities but negative in one, it was considered a false negative for the latter modality. In addition, we conducted receiver operating characteristic (ROC) analyses to determine the AUC for each test. All statistical analyses were performed using R software version 4.5.0. We used the “ROCit” package to calculate AUCs. The AUC of different tests was compared using the DeLong test.¹⁶ This study was reported in accordance with the STARD 2015 (Standards for Reporting of Diagnostic Accuracy Studies) guideline.

Standard Protocol Approvals, Registrations, and Patient Consents

This retrospective study was approved by the University of California, San Diego Institutional Review Board (UCSD IRB #811288). The requirement for informed consent was waived by the UCSD IRB because of the study’s retrospective design and use of deidentified data. No animal or identifiable human patient data were included. This study was not a registered clinical trial, as it involved retrospective analysis of existing patient records.

Data Availability

Anonymized individual participant data used for this study will be made available by request from any qualified investigator. The shared data will include deidentified demographic, clinical, and imaging variables used in the analysis. Investigators requesting access must submit a brief research proposal outlining the intended use of the data, which will be reviewed and approved by the study team. Data will be shared through secure, encrypted transfer after approval and completion of a data use agreement.

Results

Of the 469 patients with ischemic stroke evaluated at our center during the study period, 339 were excluded from the final analysis. The remaining 130 patients met ESUS criteria and had sufficient multimodal imaging to be included in the final cohort (Figure). Among these 130 patients, 91 tested

positive for a RLS by at least 1 modality and 66 were ultimately confirmed to have a PFO based on a second concordant positive test.

The mean age of the cohort was 59 years (± 16.8), and 58% were male (75/130). Common vascular risk factors included hyperlipidemia (46.1%), hypertension (35.4%), and diabetes mellitus (13.1%). Additional baseline characteristics are presented in Table 1. The median Risk of Paradoxical Embolism score was 5 (IQR: 3–7), suggesting moderate likelihood of PFO-related stroke across the group.¹⁷ Tables 2 and 3 present the diagnostic performance of each test.

TCD was performed in all 130 patients and exhibited the highest sensitivity of 100% (95% CI: 94.5–100) and a specificity of 75% (95% CI: 62.6–85.0). The test yielded a NPV of 100% (95% CI 92.6–100), a PPV of 80.4% (95% CI 72.9–86.3), and an overall accuracy of 87.6% (95% CI 80.7–92.8). When we excluded those patients with a mild PFO ($n = 5$) and those with extracardiac shunts ($n = 16$), the specificity of TCD increased to 81.1% (95% CI 68.6–90.7) and overall accuracy increased to 90.8% (95% CI 83.7–95.5).

TTE was performed in 126 patients, achieving a sensitivity of 83.3% (95% CI: 72.1–91.3) and a specificity of 93.3% (95% CI: 83.8–98.1). The NPV was 83.5% (95% CI 74.7–89.7), the PPV was 93.2% (95% CI 84.1–97.2), and the overall accuracy was 88.1% (95% CI 81.1–93.1).

TEE was performed in 81 patients, demonstrating a sensitivity of 95.1% (95% CI: 83.4–99.4), specificity of 87.5% (95% CI: 73.2–95.8), NPV of 94.5% (95% CI 81.8–98.5), PPV of 88.6% (95% CI 77.4–94.6), and the highest overall accuracy among the 4 modalities at 91.3% (95% CI 83.0–96.4).

Figure Process of Patient Inclusion and Exclusion From the Study

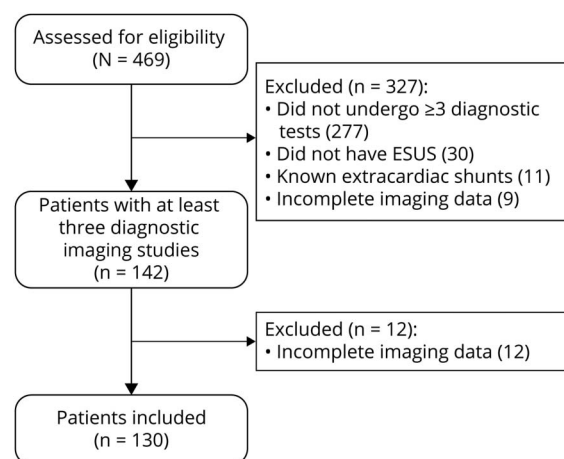


Table 1 Baseline Patient Characteristics

Clinical characteristics	N (%)
Sample size	130
Age (y) mean ± SD	59 ± 16.8
Sex (male/female)	75/55
Current smoking, (%)	24 (18.4)
Hypertension, (%)	46 (35.4)
Hyperlipidemia, (%)	60 (46.1)
Atrial fibrillation, (%)	17 (13.1)
Diabetes mellitus, (%)	17 (13.1)
COPD/OSA, (%)	12 (9.2)
History of TIA, (%)	37 (28.6)
History of malignancy, (%)	21 (16.1)
History of hypercoagulation, (%)	11 (8.4)
RoPE score, median (IQR)	5 (3–7)
Patients with confirmed PFO, (%)	66 (50.7)

Abbreviations: COPD = chronic obstructive pulmonary disease; OSA = obstructive sleep apnea; RoPE = Risk of Paradoxical Embolism.¹⁸

Cardiac CT was performed in 63 patients and had the lowest sensitivity of 45.8% (95% CI 25.5–67.1) but the highest specificity of 100% (95% CI 90.9–100). NPV was 64.2% (95% CI: 55.4–72.1), PPV was 100% (95% CI 71.5–100), and accuracy was 72.5% (95% CI 59.8–83.0).

When comparing the AUC of different tests, we found better diagnostic performance for TCD vs cardiac CT ($p = 0.04$), TTE vs cardiac CT ($p = 0.03$), and TEE vs cardiac CT ($p = 0.01$). However, there was no difference in diagnostic performance among TCD, TEE, and TTE (eTable 1).

Classification of Evidence

This study provides Class IV evidence that (1) TCD and TEE were highly sensitive in detecting PFO; (2) TTE and cardiac

CT were highly specific in detecting PFO; (3) TCD, TTE, and TEE had superior diagnostic performance over cardiac CT (as defined by AUC comparison analysis); and (4) there were no significant differences in overall diagnostic performance among TCD, TTE, and TEE.

Discussion

This study provides a comprehensive, head-to-head comparison of 4 commonly used imaging modalities for RLS detection in patients with ESUS.

While many earlier studies establishing the role of PFO used TEE alone as the primary diagnostic reference, our investigation used a more stringent, multimodality reference standard to establish the highest certainty for both the anatomical presence and the functional significance of the shunt. Specifically, the gold standard for confirming a PFO in our analysis required a positive result on at least 2 modalities of TEE, TTE, TCD, or cardiac CT. This stringent, composite reference was necessary to accurately calculate the sensitivity, specificity, and predictive values for each of the 4 individual modalities, ensuring that our performance metrics are grounded in the highest level of diagnostic confidence. Crucially, the diagnostic performance metrics derived from this strict reference demonstrate that no single noninvasive test is perfect, thereby validating our final conclusion that a staged, multimodality approach is the most effective and resource-efficient clinical strategy for PFO detection.

TCD demonstrated exceptional sensitivity (100%) and NPV (100%) in our cohort, underscoring its value as a highly sensitive and readily deployable screening tool for detecting RLS, particularly in acute or inpatient settings. These findings are consistent with previous meta-analyses reporting pooled sensitivity of 97% and specificity of 93% for TCD, with diagnostic accuracy approaching 96% when Valsalva maneuvers are effectively performed.^{8,13,18–20} However, the specificity of TCD in our study was lower than anticipated at 75%, likely because of false positives stemming from extracardiac shunts—such as pulmonary arteriovenous malformations—and small PFOs that may go undetected by other imaging modalities. This finding aligns with previous literature, including a meta-analysis by Katsanos et al., which reported TCD specificity for PFO detection ranging from 60% to 100% across multiple studies.^{8,21–25} Notably, when we excluded cases with extracardiac shunts and small PFOs, the specificity improved to 81%. The clinical significance of mild PFOs remains debated. Mild shunts, generally defined as ≤ 10 microbubbles on contrast studies, may often represent incidental findings rather than true stroke mechanisms. Previous studies and meta-analyses suggest that the risk of recurrent ischemic events and the benefit of closure are most evident in patients with moderate-to-large PFOs or those with associated atrial septal aneurysms, while small PFOs rarely confer a substantial increase in stroke risk.^{26,27} In our

Table 2 Diagnostic Test Results for Detecting Right-to-Left Shunt

Test	N, performed	TP	TN	FP	FN
TCD	130	66	48	16	0
TTE	126	55	56	4	11
TEE	81	39	35	5	2
Cardiac CT	63	11	39	0	13

Abbreviations: FN = false negative; FP = false positive; TCD = transcranial Doppler; TN = true negative; TP = true positive; TTE = transthoracic echocardiography.

Table 3 Diagnostic Value of Different Tests for Detecting Right-to-Left Shunt

Test	N, performed	Sensitivity (95% CI)	Specificity (95% CI)	NPV (95% CI)	PPV (95% CI)	Accuracy (95% CI)
TCD	130	100 (94.5–100)	75 (62.6–85.0)	100 (92.6–100)	80.4 (72.9–86.3)	87.6 (80.7–92.8)
TTE	126	83.3 (72.1–91.3)	93.3 (83.8–98.1)	83.5 (74.7–89.7)	93.2 (84.1–97.2)	88.1 (81.1–93.1)
TEE	81	95.1 (83.4–99.4)	87.5 (73.2–95.8)	94.5 (81.8–98.5)	88.6 (77.4–94.6)	91.3 (83.0–96.4)
Cardiac CT	63	45.8 (25.5–67.1)	100 (90.9–100)	64.2 (55.4–72.1)	100 (71.5–100)	72.5 (59.8–83.0)

Abbreviations: NPV = negative predictive value; PPV = positive predictive value; TCD = transcranial Doppler; TEE = transesophageal echocardiography; TTE = transthoracic echocardiography.

analysis, excluding mild PFOs improved the specificity and overall accuracy of TCD, underscoring that its diagnostic strengths are most clinically relevant when identifying moderate-to-large shunts. Although this approach reduces sensitivity for small, possibly incidental shunts, it enhances the practical utility of TCD in guiding secondary stroke prevention, where closure decisions are typically reserved for larger or high-risk PFOs. Because TCD cannot anatomically localize the source of shunting, it cannot distinguish intracardiac from extracardiac etiologies, thereby limiting its standalone diagnostic utility.^{8,21} In such cases, confirmatory imaging with TTE, TEE, or cardiac CT remains essential to verify cardiac origin.

However, this capacity to detect both intracardiac and extracardiac shunts expands the diagnostic role of TCD beyond PFO detection alone and represents an important strength of this modality in comprehensive stroke evaluation. Although the inclusion of extracardiac shunts may lower specificity when PFO is used as the reference standard, it also highlights the broader clinical utility of TCD in evaluating patients with ESUS. A positive TCD with negative cardiac imaging should, therefore, raise suspicion for an extracardiac source and prompt further diagnostic investigation.

While these findings support the role of TCD as an excellent screening test, we acknowledge that its availability is limited in many resource-constrained environments. Neurosonology laboratories and credentialed interpreters are often lacking in under-resourced hospitals, making widespread adoption challenging. Rather than assuming universal access, we propose that centers advocate for the development of training, infrastructure, and expertise to perform and interpret TCD. This approach could expand the feasibility of TCD as a first-line modality for RLS detection in patients with ESUS, particularly in hospitals where it is not yet part of routine evaluation. Despite this limitation, TCD's portability, low procedural burden, and real-time feedback make it an attractive first-line screening option, especially when echocardiography is delayed or inconclusive. In addition, the advent of robot-assisted TCD (raTCD) has the potential to expand access by allowing use in settings lacking specialized sonographers.²⁸ Recent studies also suggest that raTCD may be up to 3 times more sensitive than TTE in detecting RLS in

patients with suspected embolic stroke.^{13,27,28} TCD has been shown to be more sensitive than TTE as a first-line investigation for RLS, and growing evidence may prompt reevaluation of American Heart Association recommendations to consider TCD as a superior initial screening modality.²⁸ raTCD significantly improves sensitivity for RLS detection compared with TTE, with multicenter prospective data showing detection rates of 64% vs 20% overall and 27% vs 10% for large shunts, respectively.²⁸ This aligns with the known superiority of conventional TCD over TTE,^{13,28} and raTCD offers the added advantage of reduced operator dependency. Although not yet widely available and likely more expensive than manual TCD or TTE, raTCD systems are FDA-cleared for cerebral blood flow monitoring^{28–30}; however, specific FDA approval for RLS detection should be verified per device, and cost-effectiveness remains unproven.^{19,29,31} Accuracy seems comparable with manual TCD,^{29,30} but direct head-to-head comparisons with nonultrasound modalities such as MRI or CTA have not yet been published.

TTE demonstrated a sensitivity of 83.3%, specificity of 93.3%, PPV of 93.2%, NPV of 83.5%, and an overall diagnostic accuracy of 88.1%. These values align with existing literature, which reports sensitivity ranging from 30% to 80%, specificity near 94%, and accuracy of approximately 70%–88% for detecting RLS, particularly in cryptogenic stroke populations.^{9,28} TTE with bubble demonstrates high specificity and PPV for detecting RLS, particularly when optimized with second harmonic imaging and provocative maneuvers such as Valsalva or abdominal compression.^{32–34} Although conventional TTE has a relatively low sensitivity (~45%–50%), TTE with bubble can achieve sensitivities exceeding 85–90% and specificities between 88% and 99% when multiple contrast injections and adequate maneuvers are used.^{8,14,15,35,37} The PPV of TTE is consistently high (95%–97%) in cryptogenic stroke populations, reflecting its strong specificity and reliability in confirming clinically significant PFOs, although it may underdiagnose smaller or transient shunts in patients with suboptimal Valsalva effort.^{8,14,15,35,37} However, NPV varies widely (44%–90%), limiting its utility in ruling out small or intermittent shunts, especially when optimal technique is not used.^{12,36} Overall diagnostic accuracy of optimized TTE with bubble protocols has been reported to be up to 95%, with an AUC of 0.95 in

some meta-analyses.^{18,32} Taken together, the consistency of our findings with previous data, coupled with TTE's non-invasive profile, supports its role as a frontline diagnostic modality for PFO in ESUS, particularly in outpatient, stroke unit, and resource-limited environments.

TEE demonstrated excellent diagnostic performance, with a sensitivity of 95.1%, specificity of 87.5%, and the highest overall accuracy of 91.3%, among all modalities evaluated. These values are consistent with previous literature, which reports TEE sensitivity and specificity of approximately 96% and 93%, respectively.^{10,15} The accuracy of TEE for large RLS associated with cryptogenic stroke is around 56% per literature.¹⁵ In a prospective study by Chen et al., TEE demonstrated a PPV of 95% when compared with catheter-based or surgical confirmation. This high PPV has been reaffirmed in more recent meta-analyses and comparative studies by Liu et al. and Xu et al., confirming that TEE—particularly with contrast and provocative maneuvers—almost always corresponds to true anatomical patency.^{14,38,39} Furthermore, guidelines by the American Society of Echocardiography and expert reviews such as those by Song et al. reinforce TEE's role as the gold standard for direct septal visualization and procedural planning.^{34,40,41} The corresponding PPV of 88.6% and NPV of 94.5% in our study reinforce its reliability when both confirming and excluding clinically significant PFO. However, the slightly lower PPV observed compared with published data may reflect real-world variability stemming from procedural factors, such as operator experience, inconsistent sedation protocols, and variability in Valsalva maneuver performance. Despite its strong diagnostic profile, TEE's limitations should not be overlooked. Its invasive nature, the need for moderate sedation, and issues related to patient tolerance can reduce feasibility, particularly in the subacute stroke setting or among medically fragile patients.⁴¹⁻⁴³ These factors may also compromise imaging quality, especially when suboptimal patient cooperation results in inadequate Valsalva effort or poor esophageal windowing. Nevertheless, TEE remains indispensable when anatomical clarification is essential, especially in patients with ESUS with high pretest probability of PFO, inconclusive findings on noninvasive tests, or a history of recurrent strokes.^{8,42-44} It also plays a vital role in evaluating for coexistent cardiac or pulmonary venous anomalies, such as atrial septal aneurysms or Eustachian valve prominence, which may influence treatment decisions.^{8,42,44} Given its comprehensive structural resolution, TEE continues to be the modality of choice in complex or high-risk stroke presentations where detailed intracardiac assessment is warranted.^{8,42,44}

Cardiac CT demonstrated perfect specificity (100%) and a PPV of 100% in our cohort, indicating that when a PFO is visualized on CT, it is highly likely to be truly present. However, its diagnostic utility was limited by a markedly lower sensitivity of 45.8%, resulting in a moderate overall accuracy of 72.5%. These findings are consistent with limited

existing literature, which reports a modest sensitivity of approximately 72% and a specificity of 95% for cardiac CT in detecting potential embolic sources in cryptogenic stroke, when using TEE as the reference standard.¹⁵ Although cardiac CT can offer high specificity for PFO detection, its diagnostic performance varies significantly based on imaging protocols. When using full cardiac cycle acquisition, studies report high sensitivity and accuracy, but single-phase or prospectively gated CT protocols yield much lower sensitivity, as low as 25% in some cases.^{15,45} Older studies using multidetector CT with contrast jet criteria have shown sensitivity up to 73% and specificity near 98%, with improved sensitivity for large shunts.^{46,47} However, the American College of Cardiology and affiliated societies consider cardiac CT rarely appropriate as a first-line modality for routine PFO detection, favoring echocardiographic techniques instead.⁴⁸ Therefore, although cardiac CT may serve as a useful adjunct in select patients, particularly when TEE is contraindicated or detailed thoracic anatomy is needed, it should not replace functional imaging methods for routine RLS evaluation. The relatively poor sensitivity in our cohort underscores a fundamental limitation of cardiac CT, which is its inability to detect functional shunting under dynamic physiologic conditions.⁴⁵⁻⁴⁷ Unlike echocardiographic modalities, cardiac CT cannot simulate or capture transient RLS induced by provocative maneuvers such as the Valsalva or coughing, which are often necessary to unmask intermittent shunts. Despite these limitations, cardiac CT retains value as an adjunctive imaging modality, particularly in select clinical scenarios. It may be especially useful when echocardiographic techniques are contraindicated, poorly tolerated, or technically limited, for example, in patients with challenging acoustic windows, esophageal disease precluding TEE, or significant hemodynamic instability.⁴⁸ Moreover, cardiac CT provides excellent spatial resolution and allows for detailed anatomical assessment of the interatrial septum and surrounding thoracic structures, which may be relevant when planning interventional closure or evaluating for alternative embolic sources such as aortic arch atheroma, pulmonary pathology, or anomalous pulmonary venous return.^{47,48} In centers with advanced cardiac imaging infrastructure and experienced readers, cardiac CT can also serve as part of a multimodal diagnostic strategy to confirm structural abnormalities detected on other modalities or to provide complementary anatomical insights.⁴⁷ Although not ideal as a standalone test for PFO detection because of its low functional sensitivity, cardiac CT remains a valuable tool in the broader diagnostic and preprocedural evaluation of patients with ESUS.^{45,47,48}

In summary, our findings support a TCD-first approach for evaluating RLS in patients with ESUS. TCD proved to be a highly sensitive, noninvasive, and practical screening tool that offers strong rule-out capability for PFO. Its real-time, bedside utility and portability make it particularly valuable in acute care, inpatient, and resource-limited settings. Although TCD cannot anatomically localize a shunt and may detect

TAKE-HOME POINTS

- Transcranial Doppler (TCD) with bubble study is a highly sensitive first-line tool to screen for RLS in patients with embolic stroke of undetermined source.
- Transthoracic and transesophageal echocardiography (TEE), provide confirmatory value, with TEE offering the highest accuracy and anatomical detail.
- Cardiac CT shows excellent specificity but limited sensitivity, making it an adjunct rather than a primary diagnostic test.
- Study limitations include referral bias (all patients underwent TCD and had suspected patent foramen ovale [PFO]) and selection bias from requiring ≥ 3 diagnostic modalities, which may limit generalizability of results.
- A stepwise, multimodal approach beginning with TCD can optimize PFO detection and guide secondary stroke prevention in embolic stroke of undetermined source.

extracardiac RLS, these limitations can be addressed with confirmatory imaging, typically TTE or TEE, when results are positive. Multiple studies underscore TCD's value in this role. TCD has demonstrated a greater ability than TEE to detect RLS at rest, reinforcing its complementary role in PFO diagnosis.²³ Similarly, Droste et al. found that TCD using Echovist-300 achieved 100% sensitivity for cardiac RLS confirmed by TEE, and that detection improved with repeat testing and optimized protocols, especially for identifying clinically relevant shunts.²¹ TCD also plays a key role in grading the severity of RLS, an assessment that is not feasible with other modalities such as TTE or cardiac CT, which primarily provide structural rather than functional information.⁴⁹

In light of these findings, a stepwise, multimodal diagnostic strategy has been proposed—beginning with TCD as the initial screening tool, followed by TTE for shunt quantification and TEE for detailed anatomical characterization, particularly in patients with large bubble loads, recurrent stroke, or inconclusive noninvasive findings.^{8,12,14} TCD's sensitivity and real-time functional assessment make it an ideal first-line test, while TTE, although broadly accessible and noninvasive, may be limited by acoustic window quality and operator variability and is best used to corroborate TCD-positive cases. A TCD-first approach is most advantageous when TCD expertise is readily available, as it minimizes false negatives and efficiently rules out PFO. However, a TTE-first approach may be more practical in many centers

because TTE is universally available, and if negative, proceeding to TCD provides a sensitive adjunct to avoid missed diagnoses. TEE, despite being invasive and often requiring sedation, provides superior visualization of the interatrial septum and remains essential when structural abnormalities—such as atrial septal aneurysms or Eustachian valve prominence—are suspected. Cardiac CT, although highly specific, lacks the ability to simulate provocative maneuvers such as Valsalva, which limits its utility in detecting functional or transient right-to-left shunts. It is most appropriate in patients unable to tolerate TEE or when broader cardiothoracic anatomical evaluation is warranted. Together, these findings support a tiered diagnostic algorithm that begins with TCD, followed by selective use of TTE, TEE, or cardiac CT depending on the clinical scenario and institutional resources. Given the operator dependence of ultrasound-based techniques, ensuring standardized protocols and adequate training is essential to optimize diagnostic accuracy and reduce variability in RLS detection.

This study has several important limitations that should be considered when interpreting the findings. First, this study's retrospective design introduces potential selection bias, as clinicians may have preferentially ordered tests such as TEE for patients with a higher likelihood of PFO, possibly inflating the reported accuracy of these modalities. At the same time, this real-world practice pattern underscores a key strength of our findings: despite such variability, TCD consistently demonstrated excellent sensitivity and NPV, confirming its reliability in ruling out RLS even when applied across a broad, heterogeneous inpatient cohort.

Second, ultrasound-based techniques, particularly TTE, TEE, and TCD, are inherently operator-dependent and patient-dependent, which can affect diagnostic accuracy. Variability may arise from technical expertise and inconsistent Valsalva maneuvers, especially in sedated or uncooperative patients. Third, the sample sizes for certain modalities, especially TEE and cardiac CT, were smaller relative to TCD and TTE, limiting the statistical power to draw definitive conclusions about their comparative performance and reducing the generalizability of findings. Fourth, our study did not include formal grading of RLS across the different modalities. Grading shunt size is clinically relevant, as the sensitivity and specificity of diagnostic tools may vary depending on shunt burden and clinical context. Finally, although more than 400 patients were screened during the study period, only 130 met the inclusion criteria of undergoing at least 3 diagnostic modalities. This relatively small cohort limits statistical power and generalizability, underscoring the need for larger, prospective multicenter studies to confirm these findings.

In conclusion, this study affirms TCD as a sensitive and practical first-line tool for detecting RLS in ESUS. Although it does not provide anatomical detail, confirmatory imaging with TTE, TEE, or cardiac CT can fill that gap when needed. A TCD-first approach may streamline evaluations—especially in acute care or limited-resource settings—and optimize diagnostic yield. Future

prospective, multicenter studies with standardized protocols and long-term outcome data are essential to validate this approach and refine evidence-based stroke prevention strategies for patients with ESUS.

Author Contributions

V. Ravi: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. Y. Pirahanchi: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. S.B. Jazayeri: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. E. Rice: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. D.M. Meyer: drafting/revision of the manuscript for content, including medical writing for content. K. Agrawal: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. R. Modir: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. T. Hemmen: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. B.C. Meyer: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. R. Bavarsad Shahripour: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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